# 1 Model

## 1.1 Landscape and Roost Generation

We represent the landscape as a unit square containing $J$ discrete roosts. Each roost $j$ has location $\mathbf{x}_j = (x_j, y_j)$, carrying capacity $K_j$, and population compartments $S_j$, $I_j$, and $R_j$.

The landscape contains $C$ spatial clusters. Cluster centers are generated independently and uniformly within the unit square:

$$X_c, Y_c \overset{\text{iid}}{\sim} \text{Uniform}(0,1), \qquad c = 1, \ldots, C. \tag{1}$$

Each roost is assigned to a randomly selected cluster:

$$Z_j \sim \text{DiscreteUniform}(1, \ldots, C). \tag{2}$$

Roost coordinates are generated around their assigned cluster centers:

$$x_j = \min\left\{1, \max\left[0, X_{Z_j} + \epsilon_{x,j}\right]\right\}, \tag{3}$$

$$y_j = \min\left\{1, \max\left[0, Y_{Z_j} + \epsilon_{y,j}\right]\right\}, \tag{4}$$

where

$$\epsilon_{x,j}, \epsilon_{y,j} \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2_{\text{loc}}). \tag{5}$$

Let $p_{\text{modern}}$ denote the proportion of modern roosts. The number of modern roosts is

$$J_{\text{modern}} = \text{round}(J p_{\text{modern}}). \tag{6}$$

The mean carrying capacity of roost $j$ is

$$\mu_j = \begin{cases} 100, & \text{if roost } j \text{ is modern,} \\ 20, & \text{if roost } j \text{ is traditional.} \end{cases} \tag{7}$$

Carrying capacities are drawn from type-specific normal distributions:

$$K_j = \max\left\{K_{\min}, \text{round}(\widetilde{K}_j)\right\}, \tag{8}$$

where

$$\widetilde{K}_j \sim \mathcal{N}\left(\mu_j, (0.25\mu_j)^2\right). \tag{9}$$

## 1.2 Initial Conditions

Let $f_{\text{occ}}$ be the initial occupancy fraction and $f_{\text{inf}}$ the initial infected fraction. Initial population sizes are

$$N_j(0) = \text{round}(f_{\text{occ}} K_j), \tag{10}$$

$$I_j(0) = \text{round}(f_{\text{inf}} N_j(0)), \tag{11}$$

$$S_j(0) = N_j(0) - I_j(0), \tag{12}$$

$$R_j(0) = 0. \tag{13}$$

## 1.3 Daytime SIR Dynamics

Epidemiological transitions occur within individual roosts during the daytime. The total population of roost $j$ is

$$N_j(t) = S_j(t) + I_j(t) + R_j(t). \tag{14}$$

For $N_j(t) > 0$, the probability of infection for each susceptible bat is

$$p_{\text{inf},j}(t) = 1 - \exp\left(-\beta \frac{I_j(t)}{N_j(t)}\right), \tag{15}$$

where $\beta$ is the transmission parameter. The number of new infections is drawn from

$$B_j(t) \sim \text{Binomial}\left(S_j(t), p_{\text{inf},j}(t)\right). \tag{16}$$

The recovery probability is

$$p_{\text{rec}} = 1 - \exp(-\gamma), \tag{17}$$

where $\gamma$ is the recovery parameter. The number of recoveries is

$$G_j(t) \sim \text{Binomial}\left(I_j(t), p_{\text{rec}}\right). \tag{18}$$

Compartment counts are updated according to

$$S_j(t+1) = S_j(t) - B_j(t), \tag{19}$$
$$I_j(t+1) = I_j(t) + B_j(t) - G_j(t), \tag{20}$$
$$R_j(t+1) = R_j(t) + G_j(t). \tag{21}$$

If $N_j(t) = 0$, no infection or recovery events occur. These transitions conserve the population within each roost during the daytime.

## 1.4 Spatial Connectivity and Diffusion

The Euclidean distance between roosts $j$ and $k$ is

$$d_{jk} = \|\mathbf{x}_j - \mathbf{x}_k\|. \tag{22}$$

Roost connectivity is defined by an exponential distance-decay kernel:

$$C_{jk} = \begin{cases} \exp(-\lambda d_{jk}), & j \neq k, \\ 0, & j = k, \end{cases} \tag{23}$$

where $\lambda$ controls the rate of decay with distance.

Total connectivity and isolation are

$$c_j = \sum_{k=1}^{J} C_{jk}, \tag{24}$$
$$L_j = \frac{1}{c_j}. \tag{25}$$

The roost-specific diffusion coefficient is

$$D_j = D_{\text{base}} \exp(-\gamma_D L_j), \tag{26}$$

where $D_{\text{base}}$ is the baseline diffusion coefficient and $\gamma_D$ controls the effect of isolation on diffusion.

For a spatial location $\mathbf{x}$, let

$$d_j(\mathbf{x}) = \|\mathbf{x} - \mathbf{x}_j\|. \tag{27}$$

The distance weight assigned to roost $j$ is

$$w_j(\mathbf{x}) = \exp[-\lambda d_j(\mathbf{x})]. \tag{28}$$

The spatially varying diffusion coefficient is the weighted average

$$D(\mathbf{x}) = \frac{\sum_{j=1}^{J} w_j(\mathbf{x}) D_j}{\sum_{j=1}^{J} w_j(\mathbf{x})}. \tag{29}$$

## 1.5 Nighttime Spatial Redistribution

Let $u_q(\mathbf{x}, \tau)$ denote the spatial density of compartment $q \in \{S, I, R\}$ at location $\mathbf{x}$ and nighttime $\tau$. The initial spatial density is constructed from normalized Gaussian kernels centered on roosts:

$$u_q(\mathbf{x}, 0) = \sum_{j=1}^{J} n_{q,j} g_j(\mathbf{x}), \tag{30}$$

where $n_{S,j} = S_j$, $n_{I,j} = I_j$, and $n_{R,j} = R_j$, and

$$g_j(\mathbf{x}) = \frac{\exp\left[-\dfrac{\|\mathbf{x} - \mathbf{x}_j\|^2}{2\sigma_g^2}\right]}{\displaystyle\int_\Omega \exp\left[-\frac{\|\mathbf{z} - \mathbf{x}_j\|^2}{2\sigma_g^2}\right] d\mathbf{z}}. \tag{31}$$

Each compartment undergoes diffusion according to

$$\frac{\partial u_q}{\partial \tau} = \nabla \cdot [D(\mathbf{x}) \nabla u_q], \qquad q \in \{S, I, R\}. \tag{32}$$

We impose zero-flux boundary conditions:

$$\mathbf{n} \cdot D(\mathbf{x}) \nabla u_q = 0 \qquad \text{on } \partial\Omega, \tag{33}$$

where $\mathbf{n}$ is the outward unit normal. No epidemiological transitions or advective transport occur during the night. The diffusion equation is solved numerically on a two-dimensional grid.

## 1.6 Dawn Return and Roost Assignment

At dawn, bats are assigned from the continuous spatial density field back to discrete roosts. Let $N_j = S_j + I_j + R_j$ denote the current population of roost $j$. Roost attraction is

$$A_j = \left( \frac{N_j + \varepsilon}{\displaystyle\sum_{k=1}^{J} N_k + \varepsilon J} \right)^{\alpha} \left(1 - \frac{N_j}{K_j}\right), \tag{34}$$

where $\alpha$ controls population-dependent attraction and $\varepsilon$ is a small regularization constant.

For a bat located in grid cell $m$, let $d_{mj}$ be the distance from the cell to roost $j$. The unnormalized return weight is

$$W_{mj} = \exp(-\lambda d_{mj}) A_j. \tag{35}$$

The return probability is

$$P_{mj} = \frac{W_{mj}}{\displaystyle\sum_{k=1}^{J} W_{mk}}. \tag{36}$$

Roost assignment is stochastic and subject to carrying-capacity constraints:

$$S_j + I_j + R_j \le K_j \qquad \forall j. \tag{37}$$

Bats are assigned sequentially by compartment, with susceptible bats assigned first, followed by infectious and recovered bats. The resulting roost populations initiate the next day's SIR transitions.

## 1.7 Spatial Structure and Epidemic Intensity

For each landscape, let $r_j$ be the distance from roost $j$ to its nearest neighboring roost. The observed mean nearest-neighbor distance is

$$\bar{r}_{\text{obs}} = \frac{1}{J} \sum_{j=1}^{J} r_j. \tag{38}$$

For $J$ points in a unit-area landscape, the expected mean nearest- neighbor distance under spatial randomness is approximated by

$$\bar{r}_{\text{exp}} = \frac{1}{2\sqrt{J}}. \tag{39}$$

The Clark–Evans statistic is

$$R_{\text{CE}} = \frac{\bar{r}_{\text{obs}}}{\bar{r}_{\text{exp}}}. \tag{40}$$

Values below one indicate clustering relative to spatial randomness, whereas values above one indicate greater dispersion. We varied the proportion of modern roosts and the number of spatial clusters, then calculated the realized Clark–Evans statistic for each landscape.

Daily system-wide prevalence is

$$\pi(t) = \frac{\sum_{j=1}^{J} I_j(t)}{\sum_{j=1}^{J} N_j(t)}. \tag{41}$$

Epidemic intensity is quantified as peak prevalence:

$$\pi_{\max} = \max_{t=0,\ldots,T} \pi(t), \tag{42}$$

where $T$ is the final simulation day. For each experimental combination, peak prevalence is summarized across stochastic replicates and plotted against the proportion of modern roosts and the realized Clark–Evans statistic.